CS 506: Introduction to Quantum Computing

Error Analysis of the Quantum Fourier Transform

Scribe: Dhru Prajapati Lecture: October 21, 2026

1 Overview

The lecture revisited the Quantum Fourier Transform (QFT) and its inverse and examined what happens when a phase has more than n binary digits. The principal calculation was the probability of measuring a phase estimate after applying the inverse QFT. The topics developed on the board were:

- the inverse QFT for a phase represented exactly by n binary digits;
- phase-rotation gates R_ℓ and the difficulty of realizing very small rotations;
- the amplitudes of the inverse-QFT output for a general phase ω;
- a finite geometric series, its sine form, and the proposed probability bound $4/\pi^2$.

The proof of the no-cloning theorem was mentioned as postponed.

2 Notation and the Exact QFT

Let n be the number of qubits and set

$$N = 2^n. \tag{1}$$

We use $x \in \{0, 1, \ldots, N-1\}$ for an exactly representable computational-basis label; j for an index of a basis state in a superposition; and z for a possible measured output. In particular, these roles are distinct even when two indices range over the same set.

The QFT on N basis states is defined by

$$\mathrm{QFT}_N|x\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \mathrm{e}^{2\pi i x j/N}|j\rangle. \tag{2}$$

If the n binary digits of x are $x_1, \ldots, x_n$, then the associated exact binary fraction is

$$\omega = (0.x_1x_2\cdots x_n)_2 = \frac{x}{N}, \qquad x = (x_1x_2\cdots x_n)_2. \tag{3}$$

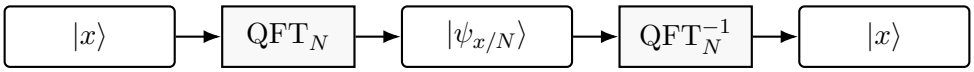
For this special phase, the state written on the board takes the form

$$\begin{aligned} |\psi_\omega\rangle &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \mathrm{e}^{2\pi i \omega j}|j\rangle \qquad (4)\\ &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \mathrm{e}^{2\pi i x j/N}|j\rangle = \mathrm{QFT}_N|x\rangle. \end{aligned}$$

Consequently,

$$\boxed{\mathrm{QFT}_N^{-1}|\psi_{x/N}\rangle = |x\rangle.} \tag{5}$$

Thus, reversing the QFT circuit and inverting its gates recovers the original computational-basis state.



3 Phase-Rotation Gates and Precision

The QFT circuit uses phase-rotation gates $R_1, R_2, \ldots, R_n$, with

$$R_\ell = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{e}^{2\pi i/2^\ell} \end{pmatrix}. \tag{6}$$

In particular,

$$R_n = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{e}^{2\pi i/2^n} \end{pmatrix}. \tag{7}$$

For large ℓ, the phase $2\pi/2^\ell$ is very small, making the corresponding rotation difficult to realize in practice. The lecturer raised the question of using rotations only up to $\ell = O(\log n)$; the constant hidden in the $O(\cdot)$ notation was noted as potentially large.

4 A Phase with More than n Binary Digits

Now consider a phase with a binary expansion that may continue beyond n digits:

$$\omega = (0.x_1x_2\cdots x_nx_{n+1}x_{n+2}\cdots)_2. \tag{8}$$

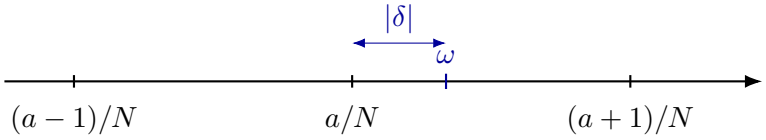
First separate the first n binary digits from the remaining binary tail, as done on the board. Let $t = (x_1x_2\cdots x_n)_2$ be the integer formed by these first n bits. Then

$$\omega = \underbrace{(0.x_1x_2\cdots x_n)_2}_{t/N} + \underbrace{(0.\underbrace{00\cdots0}_{n\text{ zeros}}x_{n+1}x_{n+2}\cdots)_2}_{\varepsilon} = \frac{t}{N} + \varepsilon, \qquad 0 \le \varepsilon < \frac{1}{N}. \tag{9}$$

The integer t is obtained by *truncation*; its residual error ε is nonnegative. For the final probability question, however, we are interested in the *nearest* n-bit estimate, which need not be the truncated value. Let a be an integer nearest to $N\omega$, and define the signed error δ by

$$\boxed{\omega = \frac{a}{N} + \delta, \qquad N\omega = a + N\delta, \qquad |\delta| \le \frac{1}{2N} = \frac{1}{2^{n+1}}.} \tag{10}$$

Here a and t are kept distinct: t denotes the first n binary digits, whereas a denotes the nearest integer used in the measurement bound. The signed error δ may be positive or negative. If $a = N$, the measured bit string is 0 modulo N because the phase is periodic.



5 Inverse-QFT Amplitudes for Arbitrary ω

We apply QFT_N^{-1} to the original phase state (4). The inverse transform acts on each computational basis state as

$$\mathrm{QFT}_N^{-1}|j\rangle = \frac{1}{\sqrt{N}} \sum_{z=0}^{N-1} \mathrm{e}^{-2\pi i j z/N} |z\rangle. \tag{11}$$

Substituting this into the expansion of $|\psi_\omega\rangle$ gives

$$\mathrm{QFT}_N^{-1}|\psi_\omega\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \mathrm{e}^{2\pi i \omega j} \left(\frac{1}{\sqrt{N}} \sum_{z=0}^{N-1} \mathrm{e}^{-2\pi i j z/N} |z\rangle \right) \tag{12}$$

$$= \frac{1}{N} \sum_{z=0}^{N-1} \left(\sum_{j=0}^{N-1} \mathrm{e}^{-2\pi i j (z - N\omega)/N} \right) |z\rangle \tag{13}$$

$$= \sum_{z=0}^{N-1} c_z |z\rangle, \tag{14}$$

where the amplitude of measuring the integer z is

$$\boxed{c_z = \frac{1}{N} \sum_{j=0}^{N-1} \mathrm{e}^{-2\pi i j (z - N\omega)/N}.} \tag{15}$$

The probability of seeing the particular result z is $\Pr[Z = z] = |c_z|^2$. It is important not to confuse the *amplitude* c_z with its squared magnitude.

Using $\omega = a/N + \delta$ from (10), we find

$$z - N\omega = (z - a) - N\delta, \tag{16}$$

$$c_z = \frac{1}{N} \sum_{j=0}^{N-1} \mathrm{e}^{-2\pi i j (z-a)/N} \, \mathrm{e}^{2\pi i j \delta}. \tag{17}$$

For the nearest output $z = a \bmod N$, the first exponential is 1 because $z - a$ is a multiple of N. Hence,

$$\boxed{c_a = \frac{1}{N} \sum_{j=0}^{N-1} \mathrm{e}^{2\pi i j \delta}, \qquad \Pr[Z = a \bmod N] = \frac{1}{N^2} \left| \sum_{j=0}^{N-1} \mathrm{e}^{2\pi i j \delta} \right|^2.} \tag{18}$$

6 Evaluating the Finite Exponential Sum

The calculation uses the following facts:

1. $\left|\mathrm{e}^{i\theta}\right|^2 = \cos^2\theta + \sin^2\theta = 1$.
2. For $r \neq 1$, a finite geometric series satisfies

$$1 + r + r^2 + \cdots + r^{N-1} = \frac{1 - r^N}{1 - r}. \tag{19}$$

3. For $|r| < 1$, the corresponding infinite series satisfies

$$1 + r + r^2 + \cdots = \frac{1}{1-r}. \tag{20}$$

For the finite sum appearing in c_a, set $r = \mathrm{e}^{2\pi i\delta}$. When $\delta \neq 0$, the geometric-series identity gives

$$\sum_{j=0}^{N-1} \mathrm{e}^{2\pi ij\delta} = \frac{1 - \mathrm{e}^{2\pi iN\delta}}{1 - \mathrm{e}^{2\pi i\delta}} \tag{21}$$

$$= \frac{\mathrm{e}^{\pi iN\delta}(\mathrm{e}^{-\pi iN\delta} - \mathrm{e}^{\pi iN\delta})}{\mathrm{e}^{\pi i\delta}(\mathrm{e}^{-\pi i\delta} - \mathrm{e}^{\pi i\delta})} \tag{22}$$

$$= \mathrm{e}^{\pi i(N-1)\delta} \frac{\sin(\pi N\delta)}{\sin(\pi\delta)}. \tag{23}$$

The complex exponential has magnitude one. Hence the probability of observing the nearest output is

$$\boxed{\Pr[Z = a \bmod N] = \frac{1}{N^2} \left| \frac{\sin(\pi N\delta)}{\sin(\pi\delta)} \right|^2.} \tag{24}$$

If $\delta = 0$, all N summands in c_a are 1. Thus the measurement gives $a \bmod N$ with probability 1.

7 Probability Bound Considered in Lecture

At the end of the lecture, the professor considered whether the probability of observing the nearest n-bit phase estimate has a constant lower bound. With the nearest-integer error satisfying

$$|\delta| \leq \frac{1}{2^{n+1}},$$

the question written next to the sine expression on the board was

$$\boxed{\Pr[Z = a \bmod 2^n] = \frac{1}{2^{2n}} \left| \frac{\sin(\pi 2^n\delta)}{\sin(\pi\delta)} \right|^2 \overset{?}{\geq} \frac{4}{\pi^2} \approx 0.4053.} \tag{25}$$

The question mark reflects the whiteboard notation: the bound was raised for consideration, but a complete proof was not shown in the photographed steps. For a nearest n-bit estimate the bound is valid; it is not generally valid for the n-bit truncation without a nearest-integer choice.

8 Conclusion

For a phase $\omega = x/2^n$ with an exact n-bit representation, the inverse QFT recovers $|x\rangle$. When ω has additional binary digits, the inverse QFT gives a distribution over possible outputs. The lecture derived the output amplitudes using a finite geometric series and expressed their probabilities in terms of a sine ratio. It concluded by asking whether the probability for the nearest n-bit estimate can be bounded below by $4/\pi^2$; the full proof was not included in the photographed board work.